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BRIEF REVIEW

Impact-Induced Muscle Damage and Contact Sports: Etiology, Effects on Neuromuscular Function and Recovery, and the Modulating Effects of Adaptation and Recovery Strategies

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Athletes involved in contact sports are habitually exposed to skeletal-muscle damage in their training and performance environments. This often leads to exercise-induced muscle damage (EIMD) resulting from repeated eccentric and/or high-intensity exercise and to impact-induced muscle damage (IIMD) resulting from collisions with opponents and the playing surface. While EIMD has been an area of extensive investigation, IIMD has received comparatively little research, with the magnitude and time frame of alterations following IIMD not presently well understood. It is currently thought that EIMD results from an overload of mechanical stress that causes ultrastructural damage to the cellular membrane constituents. Damage leads to compromised ability to produce force, which manifests immediately and persists for up to 14 d following exercise exposure. IIMD has been implicated in attenuated neuromuscular performance and recovery and in inflammatory processes, although the underlying course over time remains unclear. Exposure to EIMD leads to an adaptation to subsequent exposures, a phenomenon known as the repeated-bout effect. An analogous adaptation has been suggested to occur following IIMD; however, to date, this contention remains equivocal. While a considerable body of research has explored the efficacy of recovery strategies following EIMD, strategies promoting recovery from IIMD are limited to investigations using animal contusion models. Strategies such as cryotherapy and antioxidant supplementation that focus on attenuating the secondary inflammatory response may provide additional benefit in IIMD and are explored herein. Further research is required to first establish a model of generating IIMD and then explore broader areas around IIMD in athletic populations.

***Keywords**:* collisions, rugby, performance, fatigue

Team sports are the highest participation sports internationally. Although the physical demands of training and competition vary within and between sports, each is characterized by repeated bouts of high-intensity activity.[1,2] Unique among these activities are contact sports, such as rugby league, rugby union, American football, and Australian rules football, which are further defined by physical contact interactions between athletes. Given the extreme physical demands of these sports, athletes are regularly exposed to skeletal muscle damage in response to stressors, such as high-intensity exercise and collisions.[3,4] Muscle damage or injury can vary in severity along a continuum from mild exercise-induced muscle damage (EIMD), through varying degrees of muscular strain, to a muscular tear, significant contusion, laceration, and crush injury.[5] For athletes, muscle damage has the potential to adversely affect neuromuscular function and prolong athletic recovery.

It is well known that high-intensity and/or unaccustomed eccentric exercise causes EIMD through exposure to an inappropriate level of mechanical stress.[6] EIMD results in soreness, myofibrillar disruption, and release of intracellular proteins into the bloodstream,[7] ultimately attenuating muscle function and athletic performance.[8] The course of these alterations and strategies that may affect the course of various symptoms associated with EIMD has been areas of detailed investigation.[7,9,10]

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Additional sources of muscle tissue damage for contact-sport athletes are the contact events that occur throughout performance interactions, such as tackling, running into contact with the ball, and contact with the playing surface.[4,11] These interactions have the potential to damage underlying muscle tissue at or adjacent to the impact sites, resulting in impact-induced muscle damage (IIMD).[3] Attenuated neuromuscular function,[2] muscle soreness,[12] and increased intracellular proteins indicative of muscle damage within the blood have been observed following play.[3,4,11,13] The magnitude and timeline of these changes from IIMD are poorly understood.

Presently, the literature examining recovery from muscle damage in contact sports has been undertaken following game-play or game-simulation protocols. This typically involves evaluation of changes over time in neuromuscular function, blood parameters, subjective soreness, and mood.[1,14,15] Given that these parameters are likely to be influenced by both EIMD and IIMD, there is potential for conflation of their effects. Therefore, to explore IIMD and minimize conflation, it is pertinent to examine any potential effects in the absence of EIMD. Indeed, a previous review of musculoskeletal injuries noted that courses of progression of injury and secondary inflammatory response in EIMD and in direct muscle trauma (ie, IIMD) were divergent.[16] Furthermore, Merrick[16] noted that EIMD theory may not apply to muscle damage following blunt force trauma. These findings support the contention that IIMD is distinct from EIMD, with an independent but undefined progression of symptomatology, including altered neuromuscular function, soreness, and biochemical measures.

Considering this, the aim of this review is to critically evaluate the literature surrounding the etiology of muscle damage with

respect to EIMD and IIMD, the effects of EIMD and IIMD on neuromuscular function and recovery, and the potential adaptation to a bout of EIMD and IIMD. Consideration is also given to strategies that may attenuate IIMD, an outcome with potentially favorable consequences in situations in which recovery duration between performance bouts is limited. EIMD models have been a detailed area of investigation, and findings from this extensive body of research may provide context for the current state of IIMD research. Interventions that may acutely mitigate or attenuate EIMD and IIMD are explored. Where available, athlete data are examined in this context, and where contact-sport athlete data are lacking, animal data are explored.

Etiology of Muscle Damage

Novel or unaccustomed exercise is thought to result in EIMD via the induction of 2 damage mechanisms. The primary mechanism involves damage of the ultrastructure of the muscle tissue through an overload of mechanical stress, resulting in compromised membrane integrity and damage to the contractile apparatus (Figure 1).[16] Increased membrane permeability leads to the release of intracellular muscle-specific enzymes and proteins into the bloodstream.[17] Following primary damage, a secondary mechanism is initiated, characterized by an acute inflammatory response that results from the binding of neutrophils and macrophages to the area of damage, with subsequent release of cytokines and other pro-inflammatory factors.[7,18] The cascade of events that follows EIMD is primarily driven to remove, regenerate, and remodel the damaged tissue and facilitate skeletal muscle tissue adaptation for future exposure to potential muscle-damaging exercise.

Compression of skeletal muscle by an external stimulus leads to IIMD at and/or adjacent to the impact site.[5] Muscular damage following blunt force trauma varies in severity from mild contusion to large hematoma formation and more clinically serious complications, such as myositis ossificans (the proliferation of bone within muscle tissue).[5] Typically, this insult results in direct myofibril damage, tissue necrosis, and invasion of pro-inflammatory factors,[5,18] encompassing primary and secondary cellular injury (Figure 1).[16] Regeneration and the restoration of function typically occur over 3 overlapping stages: damage, repair, and remodeling involving the activation and interaction of immune and inflammatory systems with skeletal muscle.[18] The recovery process is somewhat mirrored in EIMD (Figure 1), although EIMD is often associated with accelerated recovery and decreased primary and secondary injury.[5,18] Although EIMD and IIMD have commonalities in regeneration and remodeling following injury, IIMD is thought to differ, with an augmented inflammatory infiltrate and subsequent secondary damage response.[16]

Appropriate healing requires the coordinated action of a number of different local and systemic factors to facilitate muscular repair following IIMD. Following a mild contusion similar to IIMD, undamaged arterioles located near the injured site dilate, increasing localized blood flow via 2 mechanisms. First, histamines are released directly from mast cells present in the damaged area, which act to increase capillary permeability via enlargement of pores within the endothelial tissue.[18] As a result, leucocyte number and levels of plasma proteins increase.[19] Concurrently, inflammatory cells release prostaglandins that are thought to sensitize local nociceptors to induce the sensation of soreness.[7] Second, local vasodilation is induced by the release of growth factors secreted by

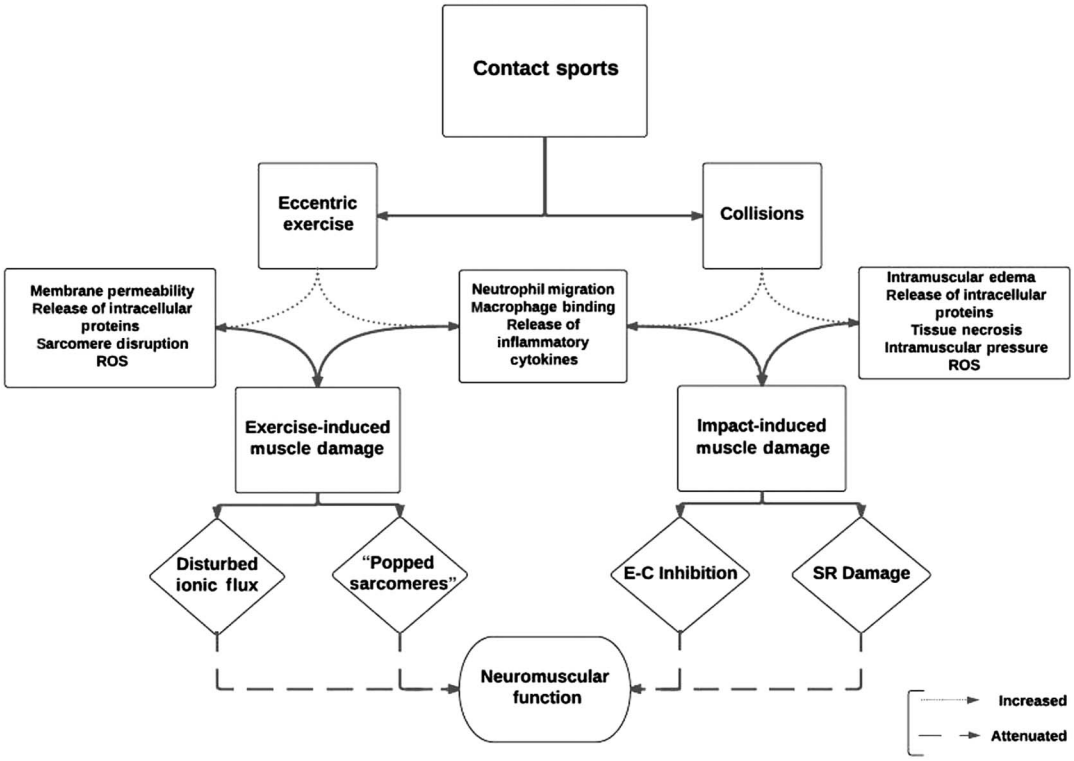

Figure 1 — Illustration of the flow of muscle damage and the respective underlying processes following exercise-induced muscle damage and impact-induced muscle damage and how these factors might attenuate neuromuscular function and performance in contact-sport athletes. ROS indicates reactive oxygen species; E-C, excitation-contraction coupling; SR, sacroplasmic reticulum.

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macrophage and endothelial tissues,[19] which in turn stimulate the nitric oxide pathway. Neutrophils, immune cells produced from bone marrow cells, migrate to the damaged area via signaling from localized myokines. Neutrophil action is implicated in free radical proliferation and secondary damage formation.[18,19] Evidence indicates that complex interactions between local and systemic factors are needed to stimulate healing and remodeling of skeletal muscle following IIMD.

To date, understanding of the effects of impact trauma on skeletal muscle tissue is drawn from the field of experimental animal contusion, in which a known mass is repeatedly dropped from a known height onto a stationary animal hind limb.[20] The histological response to this stimulus is then compared with pre-injury and/or control animal tissue. Often these investigations produce changes in the underlying tissue, including extensive myofibril necrosis, basal membrane disruption, underlying bone fracture, and incomplete recovery, associated with severe muscle injury.[20] Muscle damage and functional changes following contusion are likely to be injury severity dependent.[20] As such, generalizing these results to athletes who are habitually exposed to IIMD but report adequate recovery within the week following game play[2] should be cautioned.

Effects of Muscle Damage on Neuromuscular Function and Recovery

The decline in ability to produce muscular force in the days following muscle-damaging exercise is a consistent and robust marker of tissue damage.[7] Whereas other measures, such as soreness, systemic muscle damage marker concentrations, and cellular swelling, may not correlate well with histological evidence of damage, force loss is generally representative of EIMD severity.[6] The timeline of functional changes following EIMD typically displays a robust and consistent pattern. Strength and power losses are greatest immediately after EIMD and in the first few days, before returning to baseline over the subsequent 7 to 14 days.[6,17] Similarly, reductions in maximal voluntary torque capacity and isometric strength are typically greatest immediately following EIMD, while peak range of motion impairment is typically observed 1 to 2 days following EIMD.[17,21] Although the magnitude and progression of changes in neuromuscular function have been well characterized, the mechanism by which EIMD causes decrements in the ability to produce force is poorly understood.

Changes in neuromuscular function following EIMD have been broadly attributed to the influence of 2 mechanisms: sarcomere disruption and disturbed ionic flux (Figure 1). During active lengthening, the weakest half-sarcomeres take up most of the length change, progressively stretching beyond actin and myosin filament overlap, resulting in "popping" or disruption.[22] While most sarcomeres resume normal function at the cessation of active tension, increasing numbers become disrupted with each subsequent contraction. Structural damage to the sarcolemma and associated influx of calcium and sodium ions through stretch-activated and leak channels is another implicated mechanism for force loss.[23] Finally, accumulation of calcium-dependent proteases within the muscle has been implicated in further compromising force generation capacity.[7] These processes are thought to lead to reductions in the ability to produce force at a cellular level.

From an IIMD perspective, human and animal research has been undertaken to investigate the purported effects of high-intensity collisions and associated muscle damage on neuromuscular function indices and recovery. IIMD can disrupt capillary networks and produce intramuscular bleeding, edema, and inflammation (Figure 1).[24] These changes, along with associated rises in intramuscular pressure,[5] have been linked with displacement of actin and myosin filaments and altered sarcomere function through inhibited cross-bridge formation during muscular contractions.[24] Altered cross-bridge formation may result in attenuated force generation capacity, and this may provide a mechanism by which IIMD alters neuromuscular function and the delay in recovery.

Preliminary research by Takarada[11] established the link between collisions and muscle damage. This study examined the relationship between tackles completed during game play and systemic concentrations of markers of muscle damage: myoglobin ([Mb]) and creatine kinase ([CK]). Large and significant associations were observed between the number of tackles completed and peak [Mb] ($r = .852$, $P < .01$), and peak [CK] ($r = .922$, $P < .01$). These strong associations between physical contact interactions and muscle damage markers following contact-sport game play were further identified in several subsequent investigations.[3,4,13,15] Similarly, relationships have been observed between the number of collisions and the change in lower-limb rate of force development and peak power,[2] as well as subjective soreness,[12] following professional rugby league game play. Taken together, these data suggest a potential link between IIMD and altered muscle damage markers, soreness, and impaired athletic performance. However, beyond correlational data, there is a dearth of research investigating the effects of IIMD on exercise capacity in a controlled manner.

There are a limited number of controlled studies investigating the effects of IIMD on neuromuscular function and recovery in contact-sport populations. Studies have explored the effects of simulated collisions in the context of a small-sided game,[1] intermittent-sprint exercise,[25] and team sport conditioning circuit[26] exercise protocols. Significant decrements in functional measures have been observed, including power and mean sprint time, following IIMD, when compared with nonimpact control conditions.[1,25,26] Despite this, an inconsistent response in the change of selected blood markers of muscle damage and subjective soreness has been noted following IIMD. Johnston et al[1] identified significant increases in blood markers of muscle damage, and Singh et al[26] observed increased soreness changes when compared with a control condition. However, using similar markers and methodologies, others have observed no differences upon comparison with a control condition.[25,26] Differences in the intervention protocols and the severity of the collisions may explain this variance in response, with IIMD implemented via collisions with a tackle bag,[26] lower-body tackles performed by a research assistant,[25] or wrestling and shoulder pummels.[1] It is important to note that the inclusion of wrestling and/or high-intensity running in these studies increases the likelihood that EIMD may have resulted from the exercise protocols themselves. This may confound the results and limit the applicability of these findings to exploration of isolated IIMD on muscle damage and associated exercise performance. Caution should be exercised when generalizing based on these conclusions. In an effort to limit this effect, future research should aim to explore IIMD models in the absence of EIMD and possible muscle-damaging exercise.

From the animal research perspective, a limited number of studies have investigated neuromuscular function following IIMD. Using the drop-mass animal experimental model, Elmer et al[24] explored the effects of repeated collisions on muscular force, work, and power indices. They identified reductions in isometric force, power, and submaximal work that were significantly reduced at 1

hour (28%–33%, $P < .05$) and 48 hours (28%–38%, $P < .01$) following injury. Using a similar model, a recent investigation identified torque decrements that did not return to preinjury levels until 14 days following injury.[27] The attenuation in neuromuscular function was thought to occur due to inhibition of actomyosin cross-bridge formation caused by intracellular edema, in addition to damage to the sarcoplasmic reticulum leading to inhibited calcium release and alterations in ability to complete muscular work (Figure 1).[24]

Previous research in contact-sport populations has established the link between collisions, muscle damage, and attenuated neuromuscular function. Likewise, these studies have identified relationships between collisions and the rise in systemic muscle damage markers, further highlighting this link. Presently, the methodologies used to investigate IIMD have resulted in the possible inclusion of EIMD, and caution should be exercised when examining and attempting to draw conclusions based on this research. Further research in the animal domain has highlighted the link between IIMD and altered neuromuscular function. Attenuated neuromuscular function is thought to occur in response to increased intracellular edema resulting in inhibited cross-bridge formation and damage to the sarcoplasmic reticulum modulating calcium release.[24] Further research using models of IIMD in the absence of EIMD is needed to establish the mechanism(s) of altered neuromuscular performance in human populations, although investigations in this area may be limited by the current lack of an appropriate intervention to administer IIMD and ethical considerations. For example, it may be of interest to practitioners to see if a dose–response effect is observed following IIMD, as this could potentially influence the recovery timeline for athletes following exposure.

Adaptation to Muscle Damage

An initial bout of muscle-damaging exercise has been identified to confer a protection against the effects of a subsequent similar-intensity bout of exercise. This phenomenon is known as the repeated bout effect and has been studied extensively.[28] First identified by Highman and Altland,[29] the repeated bout effect has been noted following high-intensity exercise and eccentric muscle actions in both animal and human models.[7,28] The adaptation can persist for up to 6 months following an initial bout of muscle-damaging exercise, with observations of significant reductions in force loss, muscle damage markers, and soreness following a secondary bout.[6] Although the exact mechanism for this adaptation remains unknown, it has been broadly attributed to a number of factors, including neural, mechanical, and/or cellular changes,[30] with investigators emphasizing that adaptations in inflammatory response and extracellular matrix remodeling are likely to contribute to the repeated bout effect.[31] McHugh[30] noted that a unifying theory remains elusive, and it is likely that there are several mechanisms that either complement each other or operate independently.

Researchers have postulated that with an initial bout of IIMD a skeletal muscle adaptation occurs that affects a subsequent bout, which would be analogous to the repeated bout effect associated with EIMD.[6,28,30] Several researchers have suggested the existence of this effect, termed "contact adaptation,"[32] although these suggestions have been extrapolated from acute and longitudinal gameplay data. Hoffman et al[33] were the first to propose an adaptation between IIMD and markers of muscle damage in a study investigating the neuromuscular function and biochemical changes of athletes at an American college football game in which no change in systemic [CK] immediately following the game was observed. They noted that, as the match was the final game in a regular 12-week season, an adaptation to IIMD had occurred over the previous matches and training sessions, resulting in decreased muscle damage. An alternative hypothesis might be that there had been a repeated bout effect to EIMD induced by aspects of game play, as discussed previously. However, the investigators did observe significantly elevated [Mb] levels following the game when compared with baseline. This seemingly inconsistent finding may be due to delayed kinetics of the larger [CK] molecules escaping damaged muscle tissue into systemic circulation, in comparison to smaller [Mb] molecules that did not escape. As such, the peak of [CK] may have occurred beyond the testing time point and potentially missed in the period of data capture.

The same researchers examined changes in muscle damage markers in a longitudinal manner throughout a competitive season.[34] Testing was undertaken at 5 time points, including prior to and following the last day of the preseason training camp, and at weeks 3, 7, and 10 of the competitive season. Testing occurred at least 15 hours following the previous training session. [CK] was found to be significantly elevated following the end of the preseason camp but similar to baseline at all other time points, whereas [Mb] was similar to baseline at all time points. The authors speculated that the attenuation of [CK] following the initial training camp was evidence of adaptation to IIMD. An alternative explanation is that, following a period of relative inactivity (such as the off-season), when exposed to a period of increased training load (such as a preseason camp), athletes experienced EIMD subsequent to this increase in training load.[34] This exposure produced a repeated bout effect to EIMD which, together with repeated exposures, was observed to last the season. Interestingly, in a study investigating the longitudinal subjective soreness response to game play in elite rugby league, soreness was found to persist throughout the competitive season.[12] Upper- and lower-body soreness were found to be similar across the competitive season. These data demonstrate that athletes experience similar subjective soreness regardless of phase of season and contest the idea of adaptation to IIMD.

This highlights the equivocal state of current research exploring adaptation to IIMD. Similarly, limitations of the current research further highlight the need for future studies investigating IIMD and the importance of developing models that minimize the potential confounding effects of EIMD. Furthermore, to truly investigate any adaptive response, the inclusion of performance-based metrics is necessary, which, when combined with indices, such as subjective soreness and markers of muscle damage, may provide a holistic model of any adaptive response to IIMD. Similarly, studies exploring the relationship between these variables and the acute and longitudinal adaptive response to IIMD are limited. Further investigations can administer repeated IIMD bouts and explore the existence, magnitude, and timeline of any potential adaptive effect.

Muscle-Damage and -Recovery Strategies

Given the effects of progression and severity of muscle damage on exercise performance, a number of strategies have been explored for efficacy in promoting recovery following EIMD. These strategies have focused on modulating aspects of primary ultrastructural damage and/or the secondary inflammation response (Figure 1) to assist recovery. Strategies identified to improve aspects of recovery

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following EIMD include cryotherapy modalities,[10,35] hydrotherapies,[36,37] compression garments,[38] and dietary interventions such as antioxidant consumption[39] and protein supplementation.[39] Given the increased secondary inflammatory response associated with IIMD, strategies that preferentially target this response may provide additional benefit in aiding recovery. Cryotherapy and antioxidant supplementation are 2 strategies that may target this response and have been explored in both models of muscle damage.[35,40–44] Furthermore, due to the novel mechanism of damage associated with IIMD, prophylactic strategies, such as padding or strapping, may be effective in decreasing the magnitude of initial damage and thereby reducing the subsequent inflammatory cascade.[45,46]

Cryotherapy

Cryotherapy achieves its clinical effect by peripheral vasoconstriction and decreasing muscle temperature, resulting in decreased cellular metabolism, reduced neutrophil infiltrate, tissue necrosis, and apoptosis.[47] Intramuscular temperatures of 5°C to 15°C are believed to be optimal for reducing secondary damage,[16] although observed intramuscular temperatures typically fail to reach this threshold in healthy human models, suggesting potential perceptual effects[47] or a higher threshold for improvements than previously thought. Cold-water immersion (CWI), involving combined hydrotherapy and cryotherapy, has been shown to improve soreness when compared with passive recovery following EIMD.[44,48] A recent review noted that CWI is associated with positive effects on muscle soreness, secondary damage, and metabolite efflux, among other effects.[49]

As a means of promoting recovery, it would theoretically be beneficial to suppress cellular metabolism and limit neutrophil invasion, the inflammatory response, and the associated secondary damage following IIMD. Indeed, animal investigations have found that recovery strategies, such as cryotherapy, can attenuate inflammatory cell infiltration and oxidative stress markers and modulate edema.[50,51] Similarly, therapeutic cold application has been shown to modulate the impairment in mitochondrial function and skeletal muscle morphology seen following IIMD in animal models.[51] However, there may be conflicting functional effects, as the application of topical ice delayed regeneration and resulted in denser scar tissue formation following injury.[50] Presently, the research is limited to these few studies examining the effects of CWI in animal models of IIMD. Therefore, further research is needed to establish if commonly used strategies, such as CWI, are effective in promoting acute recovery in athletic models of IIMD.

Antioxidants

Muscle damage increases local concentrations of reactive oxygen species (ROS) that impose oxidative stress upon the tissue (Figure 1).[18] Accumulation of ROS may exacerbate damage to the ultrastructural constituents and has been implicated in the secondary damage that follows primary membrane disturbance.[10] Antioxidant compounds are known to reduce ROS and, therefore, decrease the oxidative stress imposed upon the tissue.[10,39] A number of studies have examined supplementation with antioxidant vitamins C and E to prevent or attenuate EIMD. The findings of the effects of vitamins C and E were equivocal, with observed improvements in soreness following supplementation in some studies,[10,40] whereas others observed no effect of supplementation on soreness, strength loss, or range of motion following EIMD.[41] It is important to note that antioxidant supplementation in large doses has the potential to decrease endurance training adaptations[52] and to negatively influence the cell signaling response to resistance training, resulting in decreased strength adaptations.[53] Given these findings, a shift toward examining food sources rich in antioxidant compounds known as polyphenols has been undertaken to explore their potential role in attenuating EIMD.[10,54,55] Tart cherry juice is a polyphenol-rich food source, and its supplementation has been shown to improve strength recovery, inflammatory markers, oxidative stress markers, and soreness when compared with a placebo.[54,55] Similar results were observed following supplementation with blueberries, another polyphenol-rich food source.[56] Tart cherry juice supplementation appears to be one feasible and well-supported strategy to promote recovery from EIMD by limiting secondary inflammation and damage, with blueberry supplementation showing promise as a strategy to aid recovery.

Polyphenol supplementation following IIMD has been studied in animal contusion models. Supplementation with grape seed–derived polyphenols was used following contusion injury and led to improved satellite cell response, decreased neutrophil infiltration, and decreased inflammatory cell response, suggesting accelerated recovery.[42,43] Although these findings were noteworthy, improvement in recovery of indices of neuromuscular function following supplementation was not examined. It is difficult to draw conclusions based on the limited data. Although early results are promising, more research is required to establish if the benefits of polyphenols and other antioxidant compounds translate into improved functional recovery following IIMD.

When considering the efficacy of strategies targeting the inflammatory response, a recent review has highlighted that interfering with the kinetics of inflammation negatively alters recovery from muscular injury, resulting in greater collagen deposition and smaller regenerated muscle fibers.[9] This highlights the crucial role of the acute inflammatory phase following injury in signaling subsequent sequences of regeneration. Caution should be advised when prescribing anti-inflammatory strategies following IIMD. Given this finding and the current lack of evidence for the utility of these strategies, more research is necessary to establish whether anti-inflammatory strategies improve recovery following IIMD.

Preventive Interventions

The use of preventive interventions like protective equipment has been evaluated from several perspectives, including the prevention of injuries, in a number of contact-sport cohorts. Tackles and collisions account for over 50% of injuries in adolescent rugby union athletes,[57] with body contact the predominant cause of injuries in Australian rules football.[58] In addition, the lower limb is the predominant site of injuries (35%), with muscular and ligamentous injuries accounting for 65% of all injuries in rugby union athletes.[59] Furthermore, protective equipment may prevent the incidence of soft-tissue injuries, such as contusions, strains, and sprains, whereas strapping, taping, and guards are not effective.[45] Similarly, the use of thigh protectors has been associated with a reduction in the prevalence of hematoma experienced throughout a junior Australian rules football season.[46] Although these studies did not examine the effect of prevention strategies on IIMD per se, they pointed out that protective equipment might play a role in reducing the symptoms of primary muscle damage caused by an external stimulus (Figure 1). Further research is necessary to establish whether this is a successful prevention strategy for mitigating IIMD.

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Conclusions

Athletes participating in contact sports are regularly exposed to muscle damage due to the demands of competition and training. This muscle damage can be categorized as impact induced or exercise induced, with IIMD being comparatively unexplored. The tissue response to IIMD occurs through 3 overlapping stages: damage, repair, and remodeling. These processes involve multifaceted interactions between the immune and inflammatory systems, specific cytokines, and growth factors.

There are similarities between EIMD and IIMD in the phases of progression of the primary and secondary damage response (Figure 1) and the induction of factors, such as leukocytes, macrophages, and ROS, crucial for optimal muscle fiber regeneration. The divergence between EIMD and IIMD includes the initial damage stimulus, the greater magnitude of secondary damage in IIMD, and the delayed progression of the secondary, inflammatory damage following IIMD.[8] Given these differences, it is possible that there is an independent time frame for the neuromuscular function effects and recovery following IIMD in contact-sport athletes. Similarly, there is potential for interventions that preferentially target IIMD, based on the intervention's mechanism of action and the mechanism(s) associated with IIMD.[2]

Initial research on contact-sport athletes has established strong relationships between collisions, markers of muscle damage, and functional impairment. Further controlled research has investigated IIMD in the context of exercise and identified attenuated neuromuscular function and athletic performance in the days following IIMD. Given the previously highlighted limitations of this research, the performance implications of these alterations in neuromuscular function remain to be elucidated. Animal research exploring changes in neuromuscular function has identified decrements in the ability to produce force and power in the hours and days following IIMD. It is postulated that alterations in the ability to produce force are mediated through inhibition of cross-bridge formation and muscle fiber damage secondary to altered calcium flux. However, currently, there is insufficient evidence from human models to determine whether this hypothesis is accurate or the extent of implications on functional capacity following IIMD.

Current controlled studies using contact-sport athletes have investigated IIMD in the context of high-intensity exercise. High-intensity and/or eccentrically biased exercise is likely to result in EIMD, which may lead to underlying changes in indices of recovery. This will influence IIMD data and potentially confound results and conclusions. Therefore, to understand the magnitude and timeline of IIMD, investigations are needed that examine IIMD in the absence of high-intensity and/or eccentric exercise. Future research is needed to first establish a model of administering IIMD in lieu of EIMD, which will afford investigators an opportunity to explore the functional implications and associated timeline of changes following IIMD. This model can then be used to investigate other areas of potential interest to practitioners, such as whether a dose–response effect is observed, whether a plateau exists, and adaptation to IIMD, an effect which has been discussed in the literature.[12,33,34] Furthermore, this model may be used to identify interventions and strategies that have efficacy in mitigating, treating, and managing IIMD.

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